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945 – DoD statement on the NRC report
902 – ARPA Internet Protocol policy
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616 – Latest network maps
609 – Statement of upcoming move of NIC/NLS service
590 – MULTICS address change
588 – London node is now up
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544 – Locating on-line documentation at SRI-ARC
543 – Network journal submission and delivery
518 – ARPANET accounts
511 – Enterprise phone service to NIC from ARPANET sites
510 – Request for network mailbox addresses
440 – Scheduled network software maintenance
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249 – Coordination of equipment and supplies purchase
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154 – Exposition Style
136 – Host accounting and administrative procedures
118 – Recommendations for facility documentation
95 – Distribution of NWG/RFC's through the NIC
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2a. Host Requirements

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2b. Gateway Requirements

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3a. Address Binding (ARP, RARP)

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3b. Internet Protocol over another network (encapsulation)

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893 – Trailer encapsulations
877 – Standard for the transmission of IP datagrams over public data networks

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2682 –Performance Issues in VC-Merge Capable ATM LSRs
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2642 –Cabletron's VLS Protocol Specification
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1071 –Computing the Internet checksum
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6m. Network Management (SNMP, CMOT, RMON)

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1369 –Implementation Notes and Experience for the Internet Ethernet MIB
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1351 –SNMP Administrative Model
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2674 –Definitions of Managed Objects for Bridges with Traffic Classes, Multicast Filtering and Virtual LAN Extensions
2670 –Radio Frequency (RF) Interface Management Information Base for MCNS/DOCSIS compliant RF interfaces
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2668, 2239 - Definitions of Managed Objects for IEEE 802.3 Medium Attachment Units (MAUs)
2667 –IP Tunnel MIB
2666 –Definitions of Object Identifiers for Identifying Ethernet Chip Sets
2665, 2358, 1650 - Definitions of Managed Objects for the Ethernet-like Interface Types
2662 –Definitions of Managed Objects for the ADSL Lines
2621 –RADIUS Accounting Server MIB
2620 –RADIUS Accounting Client MIB
2619 –RADIUS Authentication Server MIB
2618 –RADIUS Authentication Client MIB
2613 –Remote Network Monitoring MIB Extensions for Switched Networks Version 1.0
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2592 –Definitions of Managed Objects for the Delegation of Management Script
2591 –Definitions of Managed Objects for Scheduling Management Operations
2584 –Definitions of Managed Objects for APPN/HPR in IP Networks
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2494 –Definitions of Managed Objects for the DS0 and DS0 Bundle Interface Type
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2456 –Definitions of Managed Objects for APPN TRAPS

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2266 –Definitions of Managed Objects for IEEE 802.12 Repeater Devices
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2063 –Traffic Flow Measurement: Architecture
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1514 –Host Resources MIB
1513 –Token Ring Extensions to the Remote Network Monitoring MIB
1512, 1285 - FDDI Management Information Base
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1471 –The Definitions of Managed Objects for the Link Control Protocol of the Point-to-Point Protocol
1461 –SNMP MIB extension for Multiprotocol Interconnect over X.25
1451 –Manager-to-Manager Management Information Base
1447 –Party MIB for version 2 of the Simple Network Management Protocol (SNMPv2)
1414 –Identification MIB
1382 –SNMP MIB Extension for the X.25 Packet Layer
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1353 –Definitions of Managed Objects for Administration of SNMP Parties
1269 –Definitions of Managed Objects for the Border Gateway Protocol: Version 3
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1227 –SNMP MUX protocol and MIB
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2713 –Schema for Representing Java(tm) Objects in an LDAP Directory
2696 –LDAP Control Extension for Simple Paged Results Manipulation
2657 –LDAPv2 Client vs the Index Mesh
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2596 –Use of Language Codes in LDAP
2589 –Lightweight Directory Access Protocol (v3): Extensions for Dynamic Directory Services

2587 –Internet X.509 Public Key Infrastructure LDAPv2 Schema
2585 –Internet X.509 Public Key Infrastructure Operational Protocols: FTP and HTTP
2560 –X.509 Internet Public Key Infrastructure Online Certificate Status Protocol - OCSP
2559 –Internet X.509 Public Key Infrastructure Operational Protocols - LDAPv2
2528 –Internet X.509 Public Key Infrastructure Representation of Key Exchange Algorithm (KEA) Keys in Internet X.509 Public Key Infrastructure Certificates
2527 –Internet X.509 Public Key Infrastructure Certificate Policy and Certification Practices Framework
2511 –Internet X.509 Certificate Request Message Format
2510 –Internet X.509 Public Key Infrastructure Certificate Management Protocols
2459 –Internet X.509 Public Key Infrastructure Certificate and CRL Profile
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2307 –An Approach for Using LDAP as a Network Information Service
2294, 1836 - Representing the O/R Address hierarchy in the X.500 Directory Information Tree
2293, 1837 - Representing Tables and Subtrees in the X.500 Directory
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2247 –Using Domains in LDAP/X.500 Distinguished Names
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2148 –Deployment of the Internet White Pages Service
2120 –Managing the X.500 Root Naming Context
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1803 –Recommendations for an X.500 Production Directory Service
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1779, 1485 - A String Representation of Distinguished Names
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1684 –Introduction to White Pages Services based on X.500
1617, 1384 - Naming and Structuring Guidelines for X.500 Directory Pilots
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1608 –Representing IP Information in the X.500 Directory
1564 –DSA Metrics (OSI-DS 34 (v3))
1562 –Naming Guidelines for the AARNet X.500 Directory Service
1491 –A Survey of Advanced Usages of X.500

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1430 –A Strategic Plan for Deploying an Internet X.500 Directory Service
1373 –Portable DUAs
1309 –Technical Overview of Directory Services Using the X.500 Protocol
1308 –Executive Introduction to Directory Services Using the X.500 Protocol
1279 –X.500 and Domains
1277 –Encoding Network Addresses to Support Operation over Non-OSI Lower Layers
1276 –Replication and Distributed Operations extensions to provide an Internet Directory using X.500
1275 –Replication Requirements to provide an Internet Directory using X.500
1274 –The COSINE and Internet X.500 Schema
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1202 –Directory Assistance service
1107 –Plan for Internet directory services

6p. Information Services (HTTP, Gopher, WAIS)

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2660 –The Secure HyperText Transfer Protocol
2656 –Registration Procedures for SOIF Template Types
2655 –CIP Index Object Format for SOIF Objects
2654 –A Tagged Index Object for use in the Common Indexing Protocol
2653 –CIP Transport Protocols
2652 –MIME Object Definitions for the Common Indexing Protocol (CIP)
2651 –The Architecture of the Common Indexing Protocol (CIP)
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2518 –HTTP Extensions for Distributed Authoring -- WEBDAV
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2397 –The "data" URL scheme
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2392, 2111 - Content-ID and Message-ID Uniform Resource Locators
2388 –Returning Values from Forms: multipart/form-data
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1729 –Using the Z39.50 Information Retrieval Protocol
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1727 –A Vision of an Integrated Internet Information Service
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1614 –Network Access to Multimedia Information
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6q. Bootstrap and Configuration Protocols (BOOTP, DHCP)

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2485 –DHCP Option for The Open Group's User Authentication Protocol
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1542, 1532 - Clarifications and Extensions for the Bootstrap Protocol
1534 –Interoperation Between DHCP and BOOTP
951 – Bootstrap Protocol

6r. Real-Time Multimedia and Quality of Service (RSVP, RTP)

2719 –Framework Architecture for Signaling Transport
2705 –Media Gateway Control Protocol (MGCP) Version 1.0
2689 –Integrated Services Mappings for Low Speed Networks
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2543 –SIP: Session Initiation Protocol
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2361 –WAVE and AVI Codec Registries
2354 –Options for Repair of Streaming Media
2343 –RTP Payload Format for Bundled MPEG
2327 –SDP: Session Description Protocol
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2216 –Network Element Service Specification Template
2215 –General Characterization Parameters for Integrated Service Network Elements
2212 –Specification of Guaranteed Quality of Service
2211 –Specification of the Controlled-Load Network Element Service
2210 –The Use of RSVP with IETF Integrated Services
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1307 –Dynamically Switched Link Control Protocol
1288, 1196, 1194, 742 - The Finger User Information Protocol
1179 –Line printer daemon protocol
978 – Voice File Interchange Protocol (VFIP)
909 – Loader Debugger Protocol
891 – DCN local-network protocols
887 – Resource Location Protocol
866 – Active users
865 – Quote of the Day Protocol
864 – Character Generator Protocol
863, 348 - Discard Protocol
862, 347 - Echo Protocol
767 – Structured format for transmission of multi-media documents
759 – Internet Message Protocol
734 – SUPDUP Protocol
666 – Specification of the Unified User-Level Protocol
621 – NIC user directories at SRI ARC
569 – NETED: A Common Editor for the ARPA Network
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109 – Level III Server Protocol for the Lincoln Laboratory NIC 360/67 Host
98, 79 - Logger Protocol Proposal
29 – Response to RFC 28

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1761 –Snoop Version 2 Packet Capture File Format
496 – TNLS quick reference card is available
494 – Availability of MIX and MIXAL in the Network
488 – NLS classes at network sites
485 – MIX and MIXAL at UCSB
431 – Update on SMFS Login and Logout
411 – New MULTICS Network Software Features
409 – Tenex interface to UCSB's Simple-Minded File System
399 – SMFS Login and Logout
390 – TSO Scenario
382 – Mathematical Software on the ARPA Network
379 – Using TSO at CCN
373 – Arbitrary Character Sets
350 – User Accounts for UCSB On-Line System
345 – Interest in Mixed Integer Programming (MPSX on NIC 360/91 at CCN)
321 – CBI Networking Activity at MITRE
311 – New Console Attachments to the USCB Host
251 – Weather data
217 – Specifications changes for OLS, RJE/RJOR, and SMFS
174 – UCLA - Computer Science Graphics Overview
122 – Network specifications for UCSB's Simple-Minded File System
121 – Network on-line operators

120 – Network PL1 subprograms
119 – Network Fortran subprograms
74 – Specifications for network use of the UCSB On-Line System

8. Network Specific (also see Section 3)

8a. ARPANET

1005, 878, 851, 802 - ARPANET AHIP-E Host Access Protocol (enhanced AHIP)
852 – ARPANET short blocking feature
789 – Vulnerabilities of network control protocols: An example
745 – JANUS interface specifications
716 – Interim Revision to Appendix F of BBN 1822
704 – IMP/Host and Host/IMP Protocol change
696 – Comments on the IMP/Host and Host/IMP Protocol changes
695 – Official change in Host-Host Protocol
692 – Comments on IMP/Host Protocol changes (RFCs 687 and 690)
690 – Comments on the proposed Host/IMP Protocol changes
687 – IMP/Host and Host/IMP Protocol changes
667 – BBN host ports
660 – Some changes to the IMP and the IMP/Host interface
642 – Ready line philosophy and implementation
638, 633 - IMP/TIP preventive maintenance schedule
632 – Throughput degradations for single packet messages
627 – ASCII text file of hostnames
626 – On a possible lockup condition in IMP subnet due to message sequencing
625 – On-line hostnames service
623 – Comments on on-line host name service
622 – Scheduling IMP/TIP down time
620 – Request for monitor host table updates
619 – Mean round-trip times in the ARPANET
613 – Network connectivity: A response to RFC 603
611 – Two changes to the IMP/Host Protocol to improve user/network communications
606 – Host names on-line
594 – Speedup of Host-IMP interface
591 – Addition to the Very Distant Host specifications
568, 567 - Response to RFC 567 - cross country network bandwidth
548 – Hosts using the IMP Going Down message
547 – Change to the Very Distant Host specification
533 – Message-ID numbers
528 – Software checksumming in the IMP and network reliability
521 – Restricted use of IMP DDT
508 – Real-time data transmission on the ARPANET
476, 434 - IMP/TIP memory retrofit schedule (rev 2)
449, 442 - Current flow-control scheme for IMPSYS
447, 445 - IMP/TIP memory retrofit schedule
417 – Link usage violation
410 – Removal of the 30-Second Delay When Hosts Come Up
406 – Scheduled IMP Software Releases
395 – Switch Settings on IMPs and TIPs
394 – Two Proposed Changes to the IMP-Host Protocol
369 – Evaluation of ARPANET services January-March, 1972
335 – New Interface - IMP/360
312 – Proposed Change in IMP-to-Host Protocol

297 – TIP Message Buffers
280 – A Draft of Host Names
274 – Establishing a local guide for network usage
273, 237 - More on standard host names
271 – IMP System change notifications
270 – Correction to BBN Report No
263 – "Very Distant" Host interface
254 – Scenarios for using ARPANET computers
247 – Proffered set of standard host names
241 – Connecting computers to MLC ports
239 – Host mnemonics proposed in RFC 226 (NIC 7625)
236 – Standard host names
233 – Standardization of host call letters
230 – Toward reliable operation of minicomputer-based terminals on a TIP
229 – Standard host names
228 – Clarification
226 – Standardization of host mnemonics
218 – Changing the IMP status reporting facility
213 – IMP System change notification
209 – Host/IMP interface documentation
208 – Address tables
73, 67 - Response to NWG/RFC 67
71 – Reallocation in Case of Input Error
70 – Note on Padding
64 – Getting rid of marking
41 – IMP-IMP Teletype Communication
25 – No High Link Numbers
19 – Two protocol suggestions to reduce congestion at swap bound nodes
17 – Some questions re: Host-IMP Protocol
12 – IMP-Host interface flow diagrams
7 – Host-IMP interface
6 – Conversation with Bob Kahn

8b. Host Front End Protocols

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773 – Comments on NCP/TCP mail service transition strategy
714 – Host-Host Protocol for an ARPANET-Type Network
689 – Tenex NCP finite state machine for connections
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636 – TIP/Tenex reliability improvements
635 – Assessment of ARPANET protocols
534, 516, 512 - Lost message detection
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489 – Comment on resynchronization of connection status proposal
425 – "But my NCP costs $500 a day"
210 – Improvement of Flow Control
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129 – Request for comments on socket name structure
128 – Bytes
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72 – Proposed Moratorium on Changes to Network Protocol
68 – Comments on Memory Allocation Control Commands: CEASE, ALL, GVB, RET, and RFNM
65 – Comments on Host/Host Protocol document #1
60 – Simplified NCP Protocol
59 – Flow Control - Fixed Versus Demand Allocation
58 – Logical Message Synchronization
57, 54 - Thoughts and Reflections on NWG/RFC 54
56 – Third Level Protocol: Logger Protocol
55 – Prototypical implementation of the NCP
50, 49, 47, 45, 44, 40, 39, 38, 36, 33 - Comments on the Meyer Proposal
42 – Message Data Types
23 – Transmission of Multiple Control Messages
22 – Host-host control message formats
18 – IMP-IMP and HOST-HOST Control Links
15 – Network subsystem for time sharing hosts
11 – Implementation of the Host-Host software procedures in GORDO
9, 1 - Host software
8 – Functional specifications for the ARPA Network
5 – Decode Encode Language (DEL)
2 – Host software

8d. ARPANET Initial Connection Protocol

202 – Possible Deadlock in ICP
197 – Initial Connection Protocol - Reviewed
161 – Solution to the race condition in the ICP
151, 148, 143, 127, 123 - Comments on a proffered official ICP: RFCs 123, 127
150 – Use of IPC Facilities: A Working Paper
145 – Initial Connection Protocol Control Commands
93 – Initial Connection Protocol
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66 – NIC - third level ideas and other noise

8e. USENET

1036 –Standard for interchange of USENET messages
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1132 –Standard for the transmission of 802.2 packets over IPX networks
935 – Reliable link layer protocols
916 – Reliable Asynchronous Transfer Protocol (RATP)
914 – Thinwire protocol for connecting personal computers to the Internet
824 – CRONUS Virtual Local Network

9. Measurement

9a. General

2724 –RTFM: New Attributes for Traffic Flow Measurement
2723 –SRL: A Language for Describing Traffic Flows and Specifying Actions for Flow Groups
2722 –Traffic Flow Measurement: Architecture

2721 –RTFM: Applicability Statement
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2679 –A One-way Delay Metric for IPPM
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2330 –Framework for IP Performance Metrics
2285 –Benchmarking Terminology for LAN Switching Devices
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1262 –Guidelines for Internet Measurement Activities
557 – Revelations in network host measurements
546 – Tenex load averages for July 1973
462 – Responding to user needs
415 – Tenex bandwidth
392 – Measurement of host costs for transmitting network data
352 – TIP Site Information Form
308 – ARPANET host availability data
286 – Network Library Information System
214, 193 - Network checkpoint
198 – Site Certification - Lincoln Labs 360/67
182 – Compilation of list of relevant site reports
180 – File system questionnaire
156 – Status of the Illinois site: Response to RFC 116
153 – SRI ARC-NIC status
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106 – User/Server Site Protocol Network Host Questionnaire
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876 – Survey of SMTP implementations
848 – Who provides the "little" TCP services?
847 – Summary of Smallberg surveys
846, 845, 843, 842, 839, 838, 837, 836, 835, 834, 833, 832 - Who talks TCP? - survey of 22 February 1983
844 – Who talks ICMP, too? - Survey of 18 February 1983
787 – Connectionless data transmission survey/tutorial
565 – Storing network survey data at the datacomputer
545 – Of what quality be the UCSB resources evaluators?
530 – Report on the Survey project
523 – SURVEY is in operation again
519 – Resource evaluation
514 – Network make-work
464 – Resource notebook framework
460 – NCP survey
459 – Network questionnaires
450 – MULTICS sampling timeout change
446 – Proposal to consider a network program resource notebook

96 – An Interactive Network Experiment to Study Modes of Access the Network Information Center
90 – CCN as a Network Service Center
81 – Request for Reference Information
78 – NCP Status Report: UCSB/Rand

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1030 –On testing the NETBLT Protocol over divers networks
996 – Statistics server
618 – Few observations on NCP statistics
612, 601, 586, 579, 566, 556, 538, 522, 509, 497, 482, 455, 443, 422, 413, 400, 391, 378 - Traffic statistics (December 1973)
603, 597, 376, 370, 367, 366, 362, 353, 344, 342, 332, 330, 326, 319, 315, 306, 298, 293, 288, 287, 267, 266 - Response to RFC 597: Host status
550 – NIC NCP experiment
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255, 252, 240, 235 - Status of network hosts

10. Privacy, Security and Authentication

10a. General

2716 –PPP EAP TLS Authentication Protocol
2712 –Addition of Kerberos Cipher Suites to Transport Layer Security (TLS)
2704 –The KeyNote Trust-Management System Version 2
2693 –SPKI Certificate Theory
2692 –SPKI Requirements
2630 –Cryptographic Message Syntax
2628 –Simple Cryptographic Program Interface (Crypto API)
2627 –Key Management for Multicast: Issues and Architectures
2538 –Storing Certificates in the Domain Name System (DNS)
2537 –RSA/MD5 KEYs and SIGs in the Domain Name System (DNS)
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2504 –Users' Security Handbook
2479 –Independent Data Unit Protection Generic Security Service Application Program Interface (IDUP-GSS-API)
2478, 2078, 1508 - The Simple and Protected GSS-API Negotiation Mechanism
2444 –The One-Time-Password SASL Mechanism
2440 –OpenPGP Message Format
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2315 –PKCS 7: Cryptographic Message Syntax Version 1.5
2314 –PKCS 10: Certification Request Syntax Version 1.5
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2267 –Network Ingress Filtering: Defeating Denial of Service Attacks which employ IP Source Address Spoofing
2246 –The TLS Protocol Version 1.0
2245 –Anonymous SASL Mechanism
2222 –Simple Authentication and Security Layer (SASL)
2196, 1244 - Site Security Handbook
2179 –Network Security For Trade Shows
2094 –Group Key Management Protocol (GKMP) Architecture
2093 –Group Key Management Protocol (GKMP) Specification
2025 –The Simple Public-Key GSS-API Mechanism (SPKM)

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1948 –Defending Against Sequence Number Attacks
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1510 –The Kerberos Network Authentication Service (V5)
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1507 –DASS - Distributed Authentication Security Service
1457 –Security Label Framework for the Internet
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1423, 1115 - Privacy Enhancement for Internet Electronic Mail: Part III: Algorithms, Modes, and Identifiers
1422, 1114 - Privacy Enhancement for Internet Electronic Mail: Part II: Certificate-Based Key Management
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1355 –Privacy and Accuracy Issues in Network Information Center Databases
1281 –Guidelines for the Secure Operation of the Internet
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1108 –US Department of Defense Security Options for the Internet Protocol
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1038 –Draft revised IP security option
1004 –Distributed-protocol authentication scheme
972 – Password Generator Protocol

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2631 –Diffie-Hellman Key Agreement Method
2612 –The CAST-256 Encryption Algorithm
2286 –Test Cases for HMAC-RIPEMD160 and HMAC-RIPEMD128
2268 –A Description of the RC2(r) Encryption Algorithm
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1810 –Report on MD5 Performance
1321 –The MD5 Message-Digest Algorithm
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2709 –Security Model with Tunnel-mode IPsec for NAT Domains
2451 –The ESP CBC-Mode Cipher Algorithms
2412 –The OAKLEY Key Determination Protocol

2411 –IP Security Document Roadmap
2410 –The NULL Encryption Algorithm and Its Use With IPsec
2409 –The Internet Key Exchange (IKE)
2408 –Internet Security Association and Key Management Protocol (ISAKMP)
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2406, 1827 - IP Encapsulating Security Payload (ESP)
2405 –The ESP DES-CBC Cipher Algorithm With Explicit IV
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2402, 1826 - IP Authentication Header
2401, 1825 - Security Architecture for the Internet Protocol
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2085 –HMAC-MD5 IP Authentication with Replay Prevention
1852 –IP Authentication using Keyed SHA
1851 –The ESP Triple DES Transform
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573 – Data and file transfer: Some measurement results
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439 – PARRY encounters the DOCTOR
420 – CCA ICCC weather demo
372 – Notes on a Conversation with Bob Kahn on the ICCC
364 – Serving remote users on the ARPANET
302 – Exercising The ARPANET
231 – Service center standards for remote usage: A user's view
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89 – Some historic moments in networking
4 – Network timetable

12. Site Documentation

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13. Protocol Standards By Other Groups Of Interest To The Internet

13a. ANSI

183 – EBCDIC codes and their mapping to ASCII
20 – ASCII format for network interchange

13b. NRC

942 – Transport protocols for Department of Defense data networks
939 – Executive summary of the NRC report on transport protocols for Department of Defense data networks

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926 – Protocol for providing the connectionless mode network services
905 – ISO Transport Protocol specification ISO DP 8073
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14a. Protocol Translation and Bridges

1086 –ISO-TP0 bridge between TCP and X.25
1029 –More fault tolerant approach to address resolution for a Multi-LAN system of Ethernets

14b. Tunneling and Layering

2661 –Layer Two Tunneling Protocol "L2TP"
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1085 –ISO presentation services on top of TCP/IP based internets
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169 – Computer networks
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1606 –A Historical Perspective On The Usage Of IP Version 9
1605 –SONET to Sonnet Translation
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1437 –The Extension of MIME Content-Types to a New Medium
1313 –Today's Programming for KRFC AM 1313 Internet Talk Radio
1217 –Memo from the Consortium for Slow Commotion Research (CSCR)
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968 – Twas the night before start-up
748 – Telnet randomly-lose option
527 – ARPAWOCKY

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Appendix 2

Glossary Of Internetworking Terms And Abbreviations

TCP/IP Terminology

Like most large enterprises, TCP/IP has a language all its own. A curious blend of networking jargon, protocol names, and abbreviations, the language is both difficult to learn and difficult to remember. To outsiders, discussions among the cognoscenti sound like meaningless babble laced with acronyms at every possible opportunity. Even after a moderate amount of exposure, readers may find that specific terms are difficult to understand. The problem is compounded because some terminology is loosely defined and because the sheer volume is overwhelming.

This glossary helps solve the problem by providing short definitions for terms used throughout the Internet. It is not intended as a tutorial for beginners. Instead, we focus on providing a concise reference to make it easy for those who are generally knowledgeable about networking to look up the meaning of specific terms or acronyms quickly. Readers will find it substantially more useful as a reference after they have studied the text than before.

A Glossary of Terms and Abbreviations In Alphabetical Order

10/100 hardware
Applied to any Ethernet hardware that can operate at either 10 Mbps or 100 Mbps.

10Base2
The technical name for the original thick Ethernet.

10Base5
The technical name for thin Ethernet.

10Base-T
The technical name for twisted pair Ethernet operating at 10 Mbps.

100Base-T
The technical name for twisted pair Ethernet operating at 100 Mbps. The term *100Base-TX* is more specific.

1000Base-T
The technical name for twisted pair Ethernet operating at 1000 Mbps (1 Gbps).

127.0.0.1
The IP *loopback* address used for testing. Packets sent to this address are processed by the local protocol software without ever being sent across a network.

2X Problem
An inefficient routing situation caused by mobile IP in which a datagram crosses the global Internet twice when traveling from a computer to a mobile that is visiting a nearby network.

576
The minimum datagram size all hosts and routers must handle.

802.3
The IEEE standard for Ethernet.

822
The TCP/IP standard format for electronic mail messages. Mail experts often refer to "822 messages." The name comes from RFC 822 that contains the specification. 822 format was previously known as 733 format.

9180
The default MTU size for sending IP datagrams over an ATM network.

AAL
(*ATM Adaptation Layer*) Part of the ATM protocols. Several adaptation layers exist; AAL5 is used for data.

ABR
Either *Available Bit Rate*, an ATM designation for service that does not guarantee a rate, or *Area Border Router*, an OSPF designation for a router that communicates with another area.

ACK
Abbreviation for *acknowledgement.*

ACK implosion
A reference to a problem that can occur with a reliable multicast protocol in which many acknowledgements (ACKs) go back to the source. Most reliable multicast schemes use designated routers to aggregate ACKs.

acknowledgement
A response sent by a receiver to indicate successful reception of information. Acknowledgements may be implemented at any level including the physical level (using voltage on one or more wires to coordinate transfer), at the link level (to indicate successful transmission across a single hardware link), or at higher levels (e.g., to allow an application program at the final destination to respond to an application program at the source).

acknowledgement aggregator
Used in a reliable multicast scheme to avoid the ACK implosion problem.

active open
The operation that a client performs to establish a TCP connection with a server at a known address.

adaptive retransmission
The scheme TCP uses to make the retransmission timer track the mean round-trip time.

address
An integer value used to identify a particular computer that must appear in each packet sent to the computer.

address binding
The translation of a higher-layer address into an equivalent lower-layer address (e.g., translation of a computer's IP address to the computer's Ethernet address).

address mask
A synonym for *subnet mask.*

address resolution
Conversion of a protocol address into a corresponding physical address (e.g., conversion of an IP address into an Ethernet address). Depending on the underlying network, resolution may require broadcasting on a local network. See ARP.

administrative scoping
A scheme for limiting the propagation of multicast datagrams. Some addresses are reserved for use within a site or within an organization.

ADSL
(*Asymmetric Digital Subscriber Line*) A popular DSL variant.

Advanced Networks and Services
The company that owned and operated the Internet backbone in 1995.

agent
In network management, an agent is the server software that runs on a host or router being managed.

AH
(*Authentication Header*) A header used by IPsec to guarantee the authenticity of a datagram's source.

all routers group
The well-known IP multicast group that includes all routers on the local network.

all systems group
The well-known IP multicast group that includes all hosts and routers on the local network.

anonymous FTP
An FTP session that uses login name *anonymous* to access public files. A server that permits anonymous FTP often allows the password *guest.*

anonymous network
A synonym for *unnumbered network.*

ANS
Abbreviation for *Advanced Networks and Services.*

ANSI
(*American National Standards Institute*) A group that defines U.S. standards for the information processing industry. ANSI participates in defining network protocol standards.

ANSNET
The Wide Area Network that formed the Internet backbone until 1995.

anycast
An address form introduced with IPv6 in which a datagram sent to the address can be routed to any of a set of computers. An *anycast address* is called a *cluster address.*

API
(*Application Program Interface*) The specification of the operations an application program must invoke to communicate over a network. The socket API is the most popular for internet communication.

application gateway
An application program that connects two or more heterogeneous systems and translates among them. E-mail gateways are especially popular.

application-server paradigm
A synonym for *client-server* paradigm.

area
In OSPF, a group of routers that exchange routing information.

area manager
A person in charge of an IETF area. The set of area managers form the IESG.

ARP
(*Address Resolution Protocol*) The TCP/IP protocol used to dynamically bind a high-level IP Address to a low-level physical hardware address. ARP is used across a single physical network and is limited to networks that support hardware broadcast.

ARPA
(*Advanced Research Projects Agency*) The government agency that funded the ARPANET, and later, the global Internet. The group within ARPA with responsibility for the ARPANET was IPTO (*Information Processing Techniques Office*), later called ISTO (*Information Systems Technology Office*). ARPA was named *DARPA* for many years.

ARPANET
A pioneering long haul network funded by ARPA (later DARPA) and built by BBN. It served from 1969 through 1990 as the basis for early networking research and as a central backbone during development of the Internet. The ARPANET consisted of individual packet switching nodes interconnected by leased lines.

ARQ
(*Automatic Repeat reQuest*) Any protocol that uses positive and negative acknowledgements with retransmission techniques to ensure reliability. The sender automatically repeats the request if it does not receive an answer.

AS
(*Autonomous System*) A collection of routers and networks that fall under one administrative entity and cooperate closely to propagate network reachability (and routing) information among themselves using an interior gateway protocol of their choice. Routers within an autonomous system have a high degree of trust. Before two autonomous systems can communicate, one router in each system sends reachability information to a router in the other.

ASN.1
(*Abstract Syntax Notation.1*) The ISO presentation standard protocol used by SNMP to represent messages.

Assigned Numbers
The RFC document that specifies (usually numeric) values used by TCP/IP protocols.

ATM

(*Asynchronous Transfer Mode*) A connection-oriented network technology that uses small, fixed-size cells at the lowest layer. ATM has the potential advantage of being able to support voice, video, and data with a single underlying technology.

ATM Adaptation Layer (AAL)

One of several protocols defined for ATM that specifies how an application sends and receives information over an ATM network. Data transmissions use AAL5.

ATMARP

The protocol a host uses for address resolution when sending IP over an ATM network.

AUI

(*Attachment Unit Interface*) The connector used for thick-wire Ethernet.

authority zone

A part of the domain name hierarchy in which a single name server is the authority.

backbone network

Any network that forms the central interconnect for an internet. A national backbone is a WAN; a corporate backbone can be a LAN.

base64

An encoding used with MIME to send non-textual data such as a binary file through e-mail.

base header

In the proposed IPng, the required header found at the beginning of each datagram.

baseband

Characteristic of any network technology like Ethernet that uses a single carrier frequency and requires all stations attached to the network to participate in every transmission. Compare to *broadband.*

bastion host

A secure computer that forms part of a security firewall and runs applications that communicate with computers outside an organization.

baud

Literally, the number of times per second the signal can change on a transmission line. Commonly, the transmission line uses only two signal states (e.g., two voltages), making the baud rate equal to the number of bits per second that can be transferred. The underlying transmission technique may use some of the bandwidth, so it may not be the case that users experience data transfers at the line's specified bit rate. For example, because asynchronous lines require 10 bit-times to send an 8-bit character, a 9600 baud asynchronous transmission line can only send 960 characters per second.

BCP

(*Best Current Practice*) A label given to a subset of RFCs that contain recommendations from the IETF about the use, configuration, or deployment of internet technologies.

Bellman-Ford

A synonym for *distance-vector.*

Berkeley broadcast

A reference to a nonstandard IP broadcast address that uses all zeros in the host portion instead of all ones. The name arises because the technique was introduced and propagated in Berkeley's BSD UNIX.

best-effort delivery

Characteristic of network technologies that do not provide reliability at link levels. IP works well over best-effort delivery hardware because IP does not assume that the underlying network provides reliability. The UDP protocol provides best-effort delivery service to application programs.

BGP

(*Border Gateway Protocol*) The major exterior gateway protocol used in the Internet. Four major versions of BGP have appeared, with BGP-4 being the current.

big endian

A format for storage or transmission of binary data in which the most-significant byte (bit) comes first. The TCP/IP standard network byte order is big endian. Compare to *little endian.*

binary exponential backoff

A technique used to control network contention or congestion quickly. A sender doubles the amount of time it waits between each successive attempt to use the network.

BISYNC

(*BInary SYNchronous Communication*) An early, low-level protocol developed by IBM and used to transmit data across a synchronous communication link. Unlike most modern link level protocols, BISYNC is byte-oriented, meaning that it uses special characters to mark the beginning and end of frames. BISYNC is often called BSC, especially in commercial products.

BNC

The style of connector used with thin-wire Ethernet.

BOOTP

Abbreviation for *BOOTstrap Protocol,* a protocol a host uses to obtain startup information, including its IP address, from a server.

bps

(*bits per second*) A measure of the rate of data transmission.

bridge

A computer that connects two or more networks and forwards packets among them. Bridges operate at the physical network level. For example, an Ethernet bridge connects two physical Ethernet cables, and forwards from one cable to the other exactly the packets that are not local. Bridges differs from repeaters because bridges store and forward complete packets, while repeaters forward all electrical signals. Bridges differ from routers because bridges use physical addresses, while routers use IP addresses.

broadband

Characteristic of any network technology that multiplexes multiple, independent network carriers onto a single cable (usually using frequency division multiplexing). For example, a single 50 Mbps broadband cable can be divided into five 10 Mbps carriers, with each treated as an independent Ethernet. The advantage of broadband is less cable; the disadvantage is higher cost for equipment at connections. Compare to *baseband*.

broadcast

A packet delivery system that delivers a copy of a given packet to all hosts that attach to it is said to broadcast the packet. Broadcast may be implemented with hardware (e.g., as in Ethernet) or with software (e.g., IP broadcasting in the presence of subnets).

broadcast and prune

A technique used in data-driven multicast forwarding in which routers forward each datagram to each network until they learn that the network has no group members.

brouter

(*Bridging ROUTER*) A device that operates as a bridge for some protocols and as a router for others (e.g., a brouter can bridge DECNET protocols and route IP).

BSC

(*Binary Synchronous Communication*) See BISYNC.

BSD UNIX

(*Berkeley Software Distribution UNIX*) The version of UNIX released by U.C. Berkeley or one of the commercial systems derived from it. BSD UNIX was the first to include TCP/IP protocols.

care-of address

A temporary IP address used by a mobile while visiting a foreign network.

category 5 cable

A standard for wiring that is used with twisted pair Ethernet.

CBT

(*Core Based Trees*) A demand-driven multicast routing protocol that builds shared forwarding trees.

CCIRN

(*Coordinating Committee for Intercontinental Research Networking*) An international group that helps coordinate international cooperation on internetworking research and development.

CCITT

(*Consultative Committee on International Telephony and Telegraphy*) The former name of International Telecommunications Union.

CDDI

(*Copper Distributed Data Interface*) An adaptation of the FDDI network technology for use over copper wires.

cell

A small, fixed-size packet. The fixed size makes hardware optimization possible. Cells are often associated with ATM networks in which a cell contains 48 octets of data and 5 octets of header.

cell tax

A reference to the 10% header overhead imposed by ATM.

CGI

(*Common Gateway Interface*) A technology a server uses to create a Web page dynamically when the request arrives.

checksum

A small, integer value computed from a sequence of octets by treating them as integers and computing the sum. A checksum is used to detect errors that result when the sequence of octets is transmitted from one machine to another. Typically, protocol software computes a checksum and appends it to a packet when transmitting. Upon reception, the protocol software verifies the contents of the packet by recomputing the checksum and comparing to the value sent. Many TCP/IP protocols use a 16-bit checksum computed with one's complement arithmetic, with all integer fields in the packet stored in network byte order.

CIDR

(*Classless Inter-Domain Routing*) The standard that specifies the details of both classless addressing and an associated routing scheme.

CL

See *connectionless service*.

class of address

The category of an IP address. The class of an address determines the location of the boundary between network prefix and host suffix.

classful addressing

The original IPv4 addressing scheme in which host addresses were divided into three classes: A, B, and C.

classless addressing
An extension of the original IPv4 addressing scheme that ignores the original class boundaries. Classless addressing was motivated by the problem of address space exhaustion.

client-server
The model of interaction in a distributed system in which a program at one site sends a request to a program at another site and awaits a response. The requesting program is called a client; the program satisfying the request is called the server. It is usually easier to build client software than server software.

closed window
A situation in TCP where a receiver has sent a window advertisement of zero because no additional buffer space is available. The sending TCP cannot transmit additional data until the receiver opens the window.

cluster address
The term originally used for *anycast* address.

CO
See *connection-oriented service.*

codec
(*coder/decoder*) A hardware device used to convert between an analog audio signal and a stream of digital values.

congestion
A situation in which traffic (temporarily) exceeds the capacity of networks or routers. TCP includes a congestion control mechanism that allows it to back off when the internet becomes congested.

connection
An abstraction provided by protocol software. TCP provides a connection from an application on one computer to an application on another.

connection-oriented service
Characteristic of the service offered by any technology that requires communicating entities to establish a connection before sending data. TCP provides connection-oriented service as does ATM hardware.

connectionless service
Characteristic of any packet delivery service that treats each packet or datagram as a separate entity and allows communicating entities to transmit data before establishing communication. Each packet carries a destination address to identify the intended recipient. Most network hardware, the Internet Protocol (IP), and the User Datagram Protocol (UDP) provide connectionless service.

COPS
(*Common Open Policy Service*) A protocol used with RSVP to verify whether a request meets policy constraints.

core architecture
Characteristic of an internet architecture that has a central routing system surrounded by local routing systems. The original Internet had a single backbone network, and used a core architecture. As ISPs developed backbone systems, the Internet moved away from a single core.

count to infinity
A popular synonym for the *slow convergence* problem.

CRC
(*Cyclic Redundancy Code*) A small, integer value computed from a sequence of octets used to detect errors that result when the sequence of octets is transmitted from one machine to another. Typically, packet switching network hardware computes a CRC and appends it to a packet when transmitting. Upon reception, the hardware verifies the contents of the packet by recomputing the CRC and comparing it to the value sent. Although more expensive to compute, a CRC detects more errors than a checksum that uses additive methods.

CR-LF
(*Carriage Return - Line Feed*) A two-character sequence used to terminate text lines in application-layer protocols such as TELNET and SMTP.

CSMA/CD
(*Carrier Sense Multiple Access with Collision Detection*) A characteristic of network hardware that operates by allowing multiple stations to contend for access to a transmission medium by listening to see if the medium is idle, and a mechanism that allows the hardware to detect when two stations simultaneously attempt transmission. Ethernet uses CSMA/CD.

CSU/DSU
(*Channel Service Unit/Data Service Unit*) An electronic device that connects a computer or router to a digital circuit leased by the telephone company. Although the device fills two rolls, it usually consists of a single physical piece of hardware.

cumulative acknowledgement
An alternative to the selective acknowledgements used by TCP. A cumulative acknowledgement reports all data that has been received successfully rather than each piece of data that arrives.

DARPA
(*Defense Advanced Research Projects Agency*) Former name of ARPA.

data-driven multicast
A scheme for multicast forwarding that uses the broadcast and prune approach. See *demand-driven multicast.*

datagram
See *IP datagram.*

DCE

(*Data Communications Equipment*) Term ITU protocol standards apply to switching equipment that forms a packet switched network to distinguish it from the computers or terminals that connect to the network. Also see *DTE.*

DDCMP

(*Digital Data Communication Message Protocol*) The link level protocol used in the original NSFNET backbone.

DDN

(*Defense Data Network*) The part of the Internet associated with U.S. military sites.

default route

A single entry in a list of routes that covers all destinations which are not included explicitly. The routing tables in most routers and hosts contain an entry for a default route.

delay

One of the two primary measures of a network. Delay refers to the difference between the time a bit of data is injected into a network and the time the bit exits.

delayed acknowledgement

A heuristic employed by a receiving TCP to avoid silly window syndrome.

demand-driven multicast

A scheme for multicast forwarding that requires a router to join a shared forwarding tree before deliverying packets. See *data-driven multicast.*

demultiplex

To separate from a common input into several outputs. Demultiplexing occurs at many levels. Hardware demultiplexes signals from a transmission line based on time or carrier frequency to allow multiple, simultaneous transmissions across a single physical cable. IP software demultiplexes incoming datagrams, sending each to the appropriate high-level protocol module or application program. See *multiplex.*

DHCP

(*Dynamic Host Configuration Protocol*) A protocol that a host uses to obtain all necessary configuration information including an IP address. DHCP is popular with ISPs because it allows a host to obtain a temporary IP address.

DiffServe

(*Differentiated Services*) A scheme adopted to replace the original IP type of service. DiffServe provides up to 64 possible types of service (e.g., priorities); each datagram carries a field in the header that specifies the type of service it desires.

directed broadcast address

An IP address that specifies "all hosts" on a specific network. A single copy of a directed broadcast is routed to the specified network where it is broadcast to all machines on that network.

distance-vector

A class of routing update protocols that use a distributed shortest path algorithm (SPF) in which each participating router sends its neighbors a list of networks it can reach and the distance to each network.

DNS

(*Domain Name System*) The on-line distributed database system used to map human-readable machine names into IP addresses. DNS servers throughout the connected Internet implement a hierarchical namespace that allows sites freedom in assigning machine names and addresses. DNS also supports separate mappings between mail destinations and IP addresses.

domain

A part of the DNS naming hierarchy. Syntactically, a domain name consists of a sequence of names (labels) separated by periods (dots).

dotted decimal notation

A syntactic form used to represent 32-bit binary integers that consists of four 8-bit numbers written in base 10 with periods (dots) separating them. Many TCP/IP application programs accept dotted decimal notation in place of destination machine names.

dotted hex notation

A syntactic form used to represent binary values that consists of hexadecimal values for each 8-bit quantity with dots separating them.

dotted quad notation

A syntactic form used to represent binary values that consists of hexadecimal values for each 16-bit quantity with dots separating them.

DS3

A telephony classification of speed for leased lines equivalent to approximately 45 Mbps.

DSL

(*Digital Subscriber Line*) A set of technologies used to provide high-speed data service over the copper wires that connect between telephone offices, local residences or businesses.

DTE

(*Data Terminal Equipment*) Term ITU protocol standards apply to computers and/or terminals to distinguish them from the packet switching network to which they connect. Also see *DCE*.

DVMRP

(*Distance Vector Multicast Routing Protocol*) A protocol used to propagate multicast routes.

E.164

An address format specified by ITU and used with ATM.

EACK

(*Extended ACKnowledgement*) Synonym for *SACK*.

echo request and reply

A type of message that is used to test network connectivity. The ping program uses ICMP echo request and reply messages.

EGP

(*Exterior Gateway Protocol*) A term applied to any protocol used by a router in one autonomous system to advertise network reachability to a router in another autonomous system. BGP-4 is currently the most widely used exterior gateway protocol.

EIA

(*Electronics Industry Association*) A standards organization for the electronics industry. Known for RS232C and RS422 standards that specify the electrical characteristics of interconnections between terminals and computers or between two computers.

encapsulation

The technique used by layered protocols in which a lower level protocol accepts a message from a higher level protocol and places it in the data portion of the low-level frame. Encapsulation means that datagrams traveling across a physical network have a sequence of headers in which the first header comes from the physical network frame, the next from the Internet Protocol (IP), the next from the transport protocol, and so on.

end-to-end

Characteristic of any mechanism that operates only on the original source and final destination. Applications and transport protocols like TCP are classified as end-to-end.

epoch date

A point in history chosen as the date from which time is measured. TCP/IP uses January 1, 1900, Universal Time (formerly called Greenwich Mean Time) as its epoch date. When TCP/IP programs exchange date or time of day they express time as the number of seconds past the epoch date.

ESP

(*Encapsulating Security Payload*) A packet format used by IPsec to send encrypted information.

Ethernet

A popular local area network technology invented at the Xerox Corporation Palo Alto Research Center. An Ethernet is a passive coaxial cable; the interconnections contain all active components. Ethernet is a best-effort delivery system that uses CSMA/CD technology. Xerox Corporation, Digital Equipment Corporation, and Intel Corporation developed and published the standard for 10 Mbps Ethernet. Originally, Ethernet used a coaxial cable. Later versions use a smaller coaxial cable (*thinnet*) or twisted pair cable (10Base-T).

Ethernet meltdown

An event that causes saturation or near saturation on an Ethernet. It usually results from illegal or misrouted packets, and typically lasts only a short time.

EUI-64

A 64-bit IEEE layer-2 addressing standard.

exponential backoff

See *binary exponential backoff.*

extension header

Any of the optional IPv6 headers that follows the base header.

eXternal Data Representation

See *XDR*.

extra hop problem

A routing problem in which a datagram takes an extra, unnecessary trip across a network. The problem can be difficult to detect because communication appears to work.

fair queueing

A well-known technique for controlling congestion in routers. Called "fair" because it restricts every host to an equal share of router bandwidth. Fair queueing is not completely satisfactory because it does not distinguish between small and large hosts or between hosts with a few active connections and those with many.

Fast Ethernet

A popular term for 100Base-T Ethernet.

FCCSET

(*Federal Coordinating Council for Science, Engineering, and Technology*) A government group noted for its report that called for high-speed computing and high-speed networking research.

FDDI

(*Fiber Distribution Data Interface*) A token ring network technology based on fiber optics. FDDI specifies a 100 Mbps data rate using 1300 nanometer light wavelength, and limits networks to approximately 200 km in length, with repeaters every 2 km or less.

FDM

(*Frequency Division Multiplexing*) The method of passing multiple, independent signals across a single medium by assigning each a unique carrier frequency. Hardware to combine signals is called a multiplexor; hardware to separate them is called a demultiplexor. Also see *TDM*.

file server

A process running on a computer that provides access to files on that computer to programs running on remote machines. The term is often applied loosely to computers that run file server programs.

FIN

A special TCP segment used to close a connection. Each side must send a FIN.

firewall

A configuration of routers and networks placed between an organization's internal internet and a connection to an external internet to provide security.

five-layer reference model

The protocol layering model used by TCP/IP. Although originally controversial, the success of TCP/IP has led to wide acceptance.

fixed-length subnetting

A subnet address assignment scheme in which all physical nets in an organization use the same mask. The alternative is *variable-length subnetting*.

flat namespace

Characteristic of any naming in which object names are selected from a single set of strings (e.g., street names in a typical city). Flat naming contrasts with hierarchical naming in which names are divided into subsections that correspond to the hierarchy of authority that administers them.

flow

A general term used to characterize a sequence of packets sent from a source to a destination. Some technologies define a separate flow for each pair of communicating applications, while others define a single flow to include all packets between a pair of hosts.

flow control

Control of the rate at which hosts or routers inject packets into a network or internet, usually to avoid congestion.

Ford-Fulkerson algorithm

A synonym for the distance-vector algorithm that refers to the researchers who discovered it.

forwarding

The process of accepting an incoming packet, looking up a next hop in a routing table, and sending the packet on to the next hop. IP routers perform datagram forwarding.

fragment extension header

An optional header used by IPv6 to mark a datagram as a fragment.

fragmentation

The process of dividing an IP datagram into smaller pieces when they must travel across a network that cannot handle the original datagram size. Each fragment has the same format as a datagram; fields in the IP header specify whether a datagram is a fragment, and if so, the offset of the fragment in the original datagram. IP software at the receiving end must reassemble fragments to produce the original datagram.